Customer Segmentation using K-means Clustering

Project Overview

Customer segmentation is a crucial technique used by businesses to divide customers into distinct groups based on their characteristics and behavior. This project focuses on applying the K-Means clustering algorithm to segment customers based on their demographic and spending patterns.
The primary objective of this project is to identify meaningful customer groups that can help businesses design targeted marketing strategies, improve customer satisfaction, and maximize profitability. By leveraging unsupervised machine learning techniques, we aim to uncover hidden patterns in customer data without predefined labels.

Dataset Summary

The dataset used in this project contains customer-related information such as:

- Customer ID
- Gender
- Age
- Annual Income (k$)
- Spending Score (1–100)

Among these, the most important features used for clustering are:

- **Annual Income (k$)**
- **Spending Score (1–100)**

These features help in understanding customer purchasing behavior and financial capacity. The dataset was checked for missing values and cleaned accordingly to ensure accurate analysis.

Exploratory Data Analysis using Python

Exploratory Data Analysis (EDA) was performed to understand the distribution and relationships within the dataset. Various visualization techniques were used:

Distribution Analysis

- Count plots were used to analyze gender distribution.
- Histograms were used to understand age distribution.

Relationship Analysis

- Scatter plots were used to examine the relationship between annual income and spending score.
- This visualization helped identify natural groupings in the data.

Feature Scaling

-Standardization was applied using Standard Scaler to bring all features to the same scale before applying clustering.

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

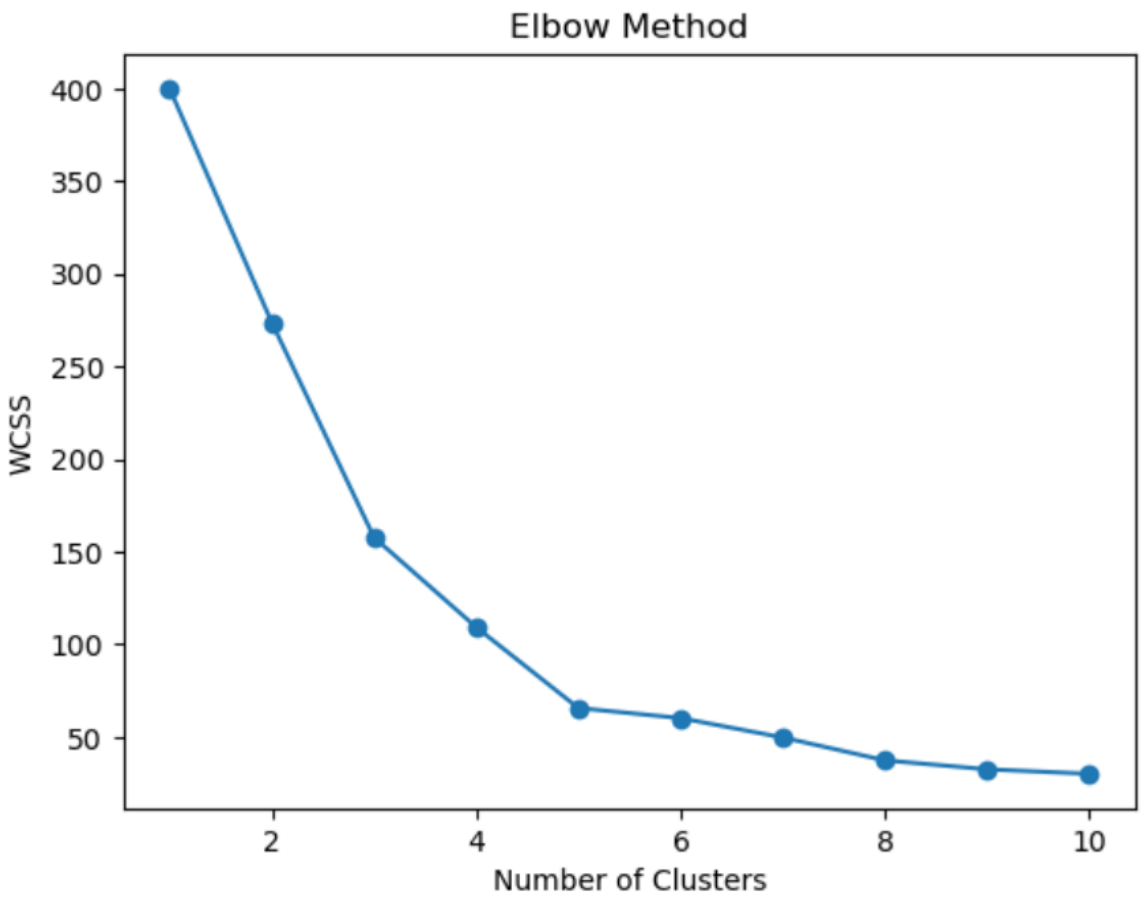
Elbow Method

- The Elbow Method was used to determine the optimal number of clusters.
- The Within-Cluster Sum of Squares (WCSS) was plotted against the number of clusters.
- The “elbow point” indicated that **5 clusters** is the optimal choice.

```
wcss = []

for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

plt.plot(range(1, 11), wcss, marker='o')
plt.title("Elbow Method")
plt.xlabel("Number of Clusters")
plt.ylabel("WCSS")
plt.show()
```



K-Means Clustering

- K-Means clustering algorithm was applied to segment customers into distinct groups.
- Each customer was assigned to a cluster based on similarity in income and spending behavior.

```
kmeans = KMeans(n_clusters=5, random_state=42, n_init=10)
df['Cluster'] = kmeans.fit_predict(X_scaled)
```

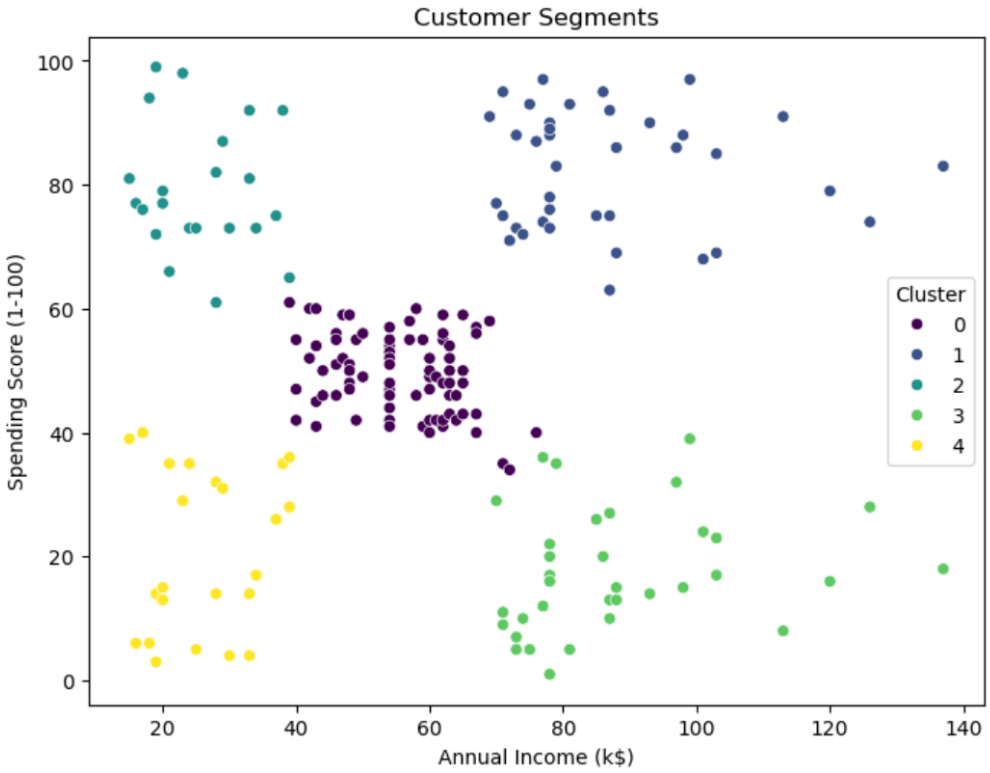
Cluster Visualization

- Scatter plots with colour coding were used to visualize different customer segments.
- Each cluster represented a unique group of customers.

```
plt.figure(figsize=(8,6))

sns.scatterplot(
    x=df['Annual Income (k$)'],
    y=df['Spending Score (1-100)'],
    hue=df['Cluster'],
    palette='viridis'
)

plt.title("Customer Segments")
plt.show()
```



Results and Insights

The clustering process revealed the following types of customer groups:

- High Income – High Spending (Premium Customers)
  High Income– Low Spending (Careful Customers)
- Low Income – High Spending (Impulsive Buyers)
- Low Income – Low Spending (Budget Customers)
- Average Income – Average Spending

These insights can help businesses:

- Target high-value customers effectively
- Design personalized marketing campaigns
- Improve customer retention strategies

Conclusion

In this project, K-Means clustering was successfully applied to segment customers based on their income and spending behavior. The use of EDA and visualization techniques helped in understanding the data and identifying meaningful patterns.

The Elbow Method ensured the selection of an optimal number of clusters, improving the quality of segmentation. The results provide valuable business insights that can be used for targeted marketing and strategic decision-making.

Overall, this project demonstrates how machine learning can be effectively used to extract actionable insights from customer data and support data-driven business strategies.

_____THANK YOU_____